Cloud Migration Design Note (DEMO DOCUMENT)

This fictional technical note describes how a small software team moves a web application from a single server to a cloud platform. It was written only for testing the DocuMind application.

Architecture Overview

The current application is a monolithic Python service that stores data in a relational database on one virtual machine. The new design splits the system into three containerised microservices: an API gateway, an authentication service and a recommendation service that uses a machine learning model.

Each service exposes a REST API that returns JSON and is deployed as a Docker container. A load balancer distributes incoming network traffic across several instances so that the platform stays available when a server fails.

Security and Data

All traffic is protected with TLS encryption, and passwords are stored as salted hashes. Two-factor authentication is required for administrators. A firewall allows only the load balancer to reach the application servers, and the database accepts connections only from the private network.

The database is migrated with an automated data pipeline that extracts the tables, validates them and loads them into a managed cloud database. Nightly backups are written to remote storage and tested every month.

Deployment and Testing

Developers use Git for version control. A continuous integration pipeline compiles the code, runs unit tests and builds the container image after every commit. Debugging in production is done through centralised logs and monitoring dashboards that track latency, errors and memory usage.